

HOERNLI, Switzerland

WMO: 066890

Lat: 47.3708N

Lon: 8.9417E

Elev: 3688

StdP: 12.84

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.7	(c) 14.5	(d) -0.8	(e) 6.0	(f) 24.2	(g) 5.6	(h) 8.4	(i) 25.8	(j) 38.9	(k) 41.4	(l) 34.0	(m) 39.3	(n) 4.1	(o) 80	(p) 0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	77.4	62.5	74.4	61.3	71.7	59.9	64.7	74.1	62.8	71.8	61.3	69.6	4.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	92.2	69.7	59.3	86.5	67.3	57.8	81.9	65.6	31.8	74.1	30.3	72.0	29.1	69.8	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 28.3	(b) 23.9	(c) 20.4	(e) 8.1	(f) 81.2	(g) 5.1	(h) 3.1	(i) 4.5	(j) 83.4	(k) 1.5	(l) 85.2	(m) -1.3	(n) 86.9	(o) -5.0	(p) 89.2		
			DB													
			WB													
				8.1	81.2	5.1	3.1	4.5	83.4	1.5	85.2	-1.3	86.9	-5.0	89.2	
				7.4	68.6	5.3	2.4	3.6	70.3	0.5	71.8	-2.5	73.1	-6.3	74.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.0	30.6	30.7	36.5	43.2	50.6	57.1	60.3	60.0	52.7	46.9	37.9	32.7
	DBStd	13.50	8.28	9.28	8.70	8.84	7.87	8.05	7.21	7.35	7.11	8.05	8.86	8.61
	HDD50	3045	601	540	421	240	91	24	7	7	53	154	371	536
	HDD65	7391	1065	960	882	656	450	259	182	192	372	561	812	1000
	CDD50	1226	0	0	4	35	109	237	325	315	133	59	9	0
	CDD65	98	0	0	0	0	3	22	35	36	2	0	0	0
	CDH74	296	0	0	0	0	3	64	114	114	1	0	0	0
	CDH80	22	0	0	0	0	0	4	10	8	0	0	0	0
(14) Wind	WSAvg	8.4	10.1	9.3	10.0	8.0	7.8	6.6	7.0	6.9	7.2	8.5	9.4	10.5
(15) Precipitation	PrecAvg	52.5	3.1	3.2	3.5	3.8	5.7	5.9	5.9	5.8	4.3	3.6	3.8	3.9
	PrecMax	66.7	7.5	8.3	9.5	7.6	11.5	9.6	11.1	10.2	9.0	6.8	8.4	8.6
	PrecMin	41.4	0.2	1.0	1.0	0.5	1.3	2.4	2.3	0.8	2.0	0.1	0.0	0.0
	PrecStd	6.6	1.9	1.7	1.9	1.9	2.5	1.8	2.4	2.3	1.8	1.7	2.1	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	52.7	57.9	66.0	72.8	80.0	81.0	80.9	71.1	67.2	60.2	54.5
		MCWB	39.3	39.1	45.3	50.5	56.9	64.6	64.5	63.1	59.8	51.9	46.7	41.3
	2%	DB	47.5	48.9	54.5	62.6	69.1	75.5	77.1	77.2	67.9	63.2	56.8	49.9
		MCWB	38.1	37.9	43.9	49.3	55.5	61.5	62.6	62.3	58.1	52.1	45.4	39.0
	5%	DB	44.4	45.9	51.6	59.2	65.7	72.4	73.9	73.9	65.4	60.2	54.0	46.3
		MCWB	37.1	36.7	42.3	47.6	53.8	60.1	61.3	61.4	56.9	51.3	43.9	37.8
	10%	DB	41.6	42.9	48.7	56.0	62.4	69.4	71.1	70.8	62.9	57.8	50.4	43.7
		MCWB	35.6	35.6	40.6	45.8	52.1	58.8	59.9	60.3	55.4	50.2	43.2	37.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	42.4	47.5	52.8	60.0	67.2	67.3	66.8	62.5	56.8	49.1	43.7
		MCDB	45.5	47.7	54.3	63.4	69.9	76.8	77.8	75.8	68.5	61.9	55.8	47.6
	2%	WB	40.5	40.2	45.4	50.5	57.2	63.6	64.4	64.5	60.0	54.5	47.2	41.7
		MCDB	44.7	45.3	52.1	60.3	66.3	73.0	74.2	73.6	65.8	59.2	53.1	46.4
	5%	WB	38.4	38.5	43.6	48.5	55.1	61.5	62.5	62.5	58.1	52.9	45.5	40.2
		MCDB	43.0	43.6	49.6	57.4	63.2	70.1	71.8	71.3	63.6	57.7	51.2	44.4
	10%	WB	36.6	36.7	41.6	46.7	53.0	59.6	60.9	60.9	56.4	51.2	43.8	38.5
		MCDB	40.7	41.5	47.2	54.6	60.7	67.7	69.3	69.1	61.6	56.2	49.4	42.5
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.6	9.1	11.0	11.7	11.7	11.6	10.5	9.2	8.8	7.8	7.7
		MCDBR	10.6	11.1	12.4	14.3	14.7	14.6	14.7	13.4	11.6	11.5	11.5	10.7
	5% WB	MCWBR	6.0	6.4	7.7	8.9	9.0	9.6	9.1	8.7	8.2	7.2	5.8	5.5
		MCWBR	9.1	10.1	11.5	13.9	14.2	14.0	13.8	12.6	10.8	9.8	10.5	9.4
(40) Clear-Sky Solar Irradiance	taub		0.260	0.281	0.312	0.344	0.357	0.363	0.365	0.358	0.334	0.310	0.283	0.256
		taud	2.465	2.420	2.414	2.372	2.374	2.388	2.373	2.408	2.461	2.522	2.537	2.476
	Ebn at Noon		269	284	290	289	287	285	283	281	279	270	258	257
		Edh at Noon	20	26	31	36	37	37	37	34	30	24	19	18
(44) All-Sky Solar Radiation	RadAvg		379	650	1033	1377	1577	1813	1757	1533	1153	728	410	312
	RadStd		42	87	140	200	158	168	160	156	118	80	51	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.48	+2.34	+1.71	+1.49	N/A	N/A	+167	+38
Regional (121 neighbors)	N/A	N/A	N/A	+1.26	+0.74	+0.59	N/A	N/A	+103	+29

Nomenclature: See separate page